

Supplementary Material

Contents

Dataset details	1
Gesture Vocabulary	1
Dataset Card and Release Format	2
EMG Signal Processing	2
Label reconstruction and quality	2
Markers2MANO Pipeline	2
Wrist Articulation Angles	3
MANO-to-UmeTrack Conversion	3
Training and implementation details	4
EMGFormer Training Details	4
Vision and Fusion Training Details	5
Extended experimental results	6
Per-Hand EMG Results	6
Per-Joint Error Analysis	6
Per-Gesture Complementarity	6
Self-Occlusion Analysis	8

Dataset details

Gesture Vocabulary

Table 5 lists the complete gesture vocabulary of *EgoEMG* with descriptions.

Table 5: Complete gesture vocabulary of *EgoEMG* (60 classes) with per-gesture descriptions.

Gesture	Description	Gesture	Description
<i>Single-hand (30 gestures)</i>		<i>Symmetric bimanual (18 gestures)</i>	
ASL1-ASL9	American Sign Language digits 1 through 9	Clap	Both hands clapping together symmetrically
Claw3	Three-finger claw pose (index + middle + ring flexed)	CrossHand	Both hands crossed fingers
Claw5	Five-finger claw pose (all fingers flexed)	CrossStretch	Hands opposite fingers and stretched
FreeAction	Free-form unconstrained hand action	FingerTipTouch	Matching fingertips of both hands touching
ILY	“I Love You” hand sign (thumb, index, pinky extended)	FistBump	Two fists bumping together
IndexBow	Index finger flexion/extension bowing	Gaming	Both hands holding a game controller
IndexMiddleClaw	Index and middle fingers clawed, others extended	HandClasp	Hands clasped together
JoystickCircle	Circular joystick manipulation on the palm with thumb	HandRub	Rubbing palms together
JoystickSlide	Linear joystick sliding on the palm with thumb	IndexTapping	Both index fingers tapping in the air
MiddleBow	Middle finger flexion/extension bowing	Kiss	Fingertips touching (thumb and middle), then separating
Nine	Hand forming number 9 (four fingers curled, index up)	PalmStack	One palm stacked on top of the other
PalmYaw	Palm facing up/down rotation about the yaw axis	Prayer	Palms pressed together in prayer position
PinchMiddle	Thumb to middle fingertip pinch	MiddleOppo	Middle and ring fingers of one hand oppose the other hand
PinkyBow	Pinky finger flexion/extension bowing	SymOpen	Both hands opening symmetrically from closed to open
Rest	Relaxed hand in neutral resting posture	SymSwing	Both hands swinging symmetrically side to side
RingAndThumb	Ring finger touching thumb tip	ThumbWrestle	Thumbs wrestling each other
RingBow	Ring finger flexion/extension bowing	Typing	Both hands typing on a virtual keyboard
Rock	“Rock on” hand sign (index and pinky extended)	raw	Unconstrained bimanual hand movement
Thumb	Thumb up	<i>Asymmetric bimanual (12 gestures)</i>	
nocontact_disperse	Fingers spread apart with palm open, no contact	FingerPullLeft	Left hand pulls right-hand fingers
_palm		FingerPullRight	Right hand pulls left-hand fingers
nocontact_free	Free-form finger motion without contact	PalmRoll	Palms facing, rolling around each other
nocontact_grab	Simulated grasping motion without contact	PinkyHook	Pinky fingers hooked together
		Squeeze	Both hands squeezing an imaginary object
		Beijing	Two hands crossing to make a “Bei” Chinese character
		Checky	Two hands with thumbs forming a “Checky” sign
		PairClaw	Both hands clawing with asymmetric finger configurations
		PairOK	Both hands forming OK signs
		Picture	Hands forming a picture frame rectangle
		PinchWring	Two hands pinching and wringing
		ThumbOppo	Thumbs in opposition at different orientations

Dataset Card and Release Format

License. The dataset will be released under CC-BY-NC 4.0 for research use, and the baseline code will be distributed under the MIT License. Third-party assets, including the MANO model (Romero, Tzionas, and Black 2017) and the training data for *markers2mano*, remain subject to their original licenses.

File schema. Data is organized by episode. Each episode is stored as a parquet file containing synchronized multimodal time series: bilateral EMG (16 channels at 2 kHz, raw and filtered), per-hand IMU, an egocentric RGB frame from a head-mounted wide-angle GoPro (1280 × 720 at 60 fps), an external RGB-D frame from a ZED 2i sensor (RGB + depth, 1280 × 720 at 30 fps), MANO pose parameters $\theta \in \mathbb{R}^{48}$ and shape parameters $\beta \in \mathbb{R}^{10}$, wrist flexion/extension and radial/ulnar deviation angles, timestamps, calibration metadata, and split assignments. Video frames are stored as MP4 files and referenced by frame index; each frame includes synchronization diagnostics (stale flag and delta_ms). Pre-cropped 256 × 256 left- and right-hand patches, generated by reprojecting mocap keypoints onto the egocentric view and extracting per-hand bounding boxes, are provided as per-episode LMDB archives to support vision-based training without requiring users to re-run the projection pipeline. Windows of 12,000 EMG samples (6 s at 2 kHz) from the target wrist’s eight channels are extracted on-the-fly during training.

Access. The dataset will be hosted with a download script and checksum verification upon publication.

EMG Signal Processing

All EMG signals are recorded at 2 kHz. The release includes both raw and filtered EMG for each channel. Filtering is applied per channel in the frequency domain via FFT mask with the following pipeline:

1. Per-channel mean subtraction (DC removal).
2. Narrow-band notch filters at 50 Hz (power line frequency) and 100 Hz (first harmonic).
3. Soft high-pass transition from 15 to 20 Hz.
4. Soft low-pass transition from 850 to 900 Hz, reaching zero gain at 900 Hz.

No amplitude normalization is applied; filtered EMG values are retained in mV. Both raw and filtered streams are preserved in the release. Training uses the filtered EMG by default.

Label reconstruction and quality

Markers2MANO Pipeline

This section details the pipeline used to recover MANO parameters from optical motion capture marker positions during *EgoEMG* data collection.

Marker Selection We select 21 vertices from the MANO mesh surface as virtual markers, distributed across the hand topology to capture wrist, finger, and palm geometry: indices 191, 88, 253, 708, 729 (wrist/palm), 144, 87, 295, 319 (thumb), 220, 365, 407, 445 (index/middle/ring), and 183, 477, 518, 556, 83, 589, 635, 673 (pinky/palm). These indices

are consistent across all MANO shape instances β , enabling direct correspondence between captured 3D marker positions and the canonical MANO mesh.

Transformer Architecture

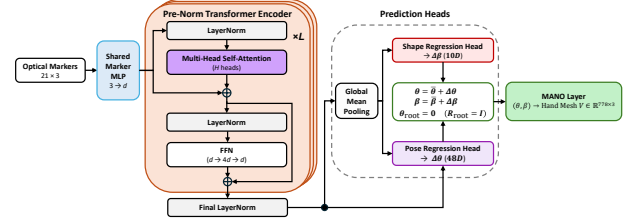


Figure 9: Architecture of *markers2mano* pipeline.

The primary inference model is a Transformer that maps 21 marker positions $\mathbf{M} \in \mathbb{R}^{21 \times 3}$ to MANO pose $\theta \in \mathbb{R}^{48}$ and shape $\beta \in \mathbb{R}^{10}$. The production configuration uses the 500M preset: embedding dimension $d = 1600$, $L = 16$ pre-norm Transformer layers, 8 attention heads, a $4d = 6400$ feed-forward width, dropout 0.1, and 499.87M trainable parameters. The architecture is shown in Figure 9. The model consists of:

- **Input embedding:** Inputs are root-centered using the wrist marker. The model accepts all 21 markers, then removes five redundant root/MCP markers (indices 1, 5, 9, 13, and 17 in marker order), retaining 16 tokens. A shared linear layer projects each 3D position to a d -dimensional token ($\mathbb{R}^3 \rightarrow \mathbb{R}^d$).
- **Transformer encoder:** L pre-norm Transformer layers, each comprising LayerNorm, multi-head self-attention, a residual connection, and a second LayerNorm-feed-forward block with a $4 \times$ expansion ratio ($d \rightarrow 4d \rightarrow d$).
- **Output heads:** After a final LayerNorm, global mean pooling feeds an MLP shape head that predicts $\Delta\beta \in \mathbb{R}^{10}$. For pose, each retained token is concatenated with the global feature and passed through a shared MLP that predicts a 6D rotation for one of the 16 MANO joints. The predicted pose and shape residuals are added to means computed from valid training labels. The heads are zero-delta initialized, and the root rotation is fixed to identity. The model does not predict global translation.
- **MANO forward pass:** The predicted θ and β are passed through the MANO layer to obtain the final mesh vertices $\mathbf{V} \in \mathbb{R}^{778 \times 3}$.

Training and Data Augmentation The model is trained on hand meshes derived from 11 public datasets using the augmentation procedure detailed below: GigaHand (Huang et al. 2024), HOT3D (Banerjee et al. 2024), ARCTIC (Fan et al. 2023), HOI4D (Liu et al. 2022), HanCo (Zimmermann, Argus, and Brox 2021), DexYCB (Yu et al. 2022), H2O (Kwon et al. 2021), InterHand2.6M (Moon et al. 2023), FreiHand (Zimmermann et al. 2019), HO3Dv3 (Hampali et al. 2020), and HO3Dv2 (Hampali et al. 2020). Their combined raw footage exceeds 195 million frames. To balance the training distribution, we assign each dataset a sampling ratio based on its scale tier: large-scale datasets (100M+ frames, e.g., GigaHand) at 0.14%; medium-scale datasets (1M+, e.g., HOT3D,

InterHand2.6M) at 1.5%; and small-scale datasets (100K+, e.g., FreiHand, HO3D) at 15–20%. This yields approximately 0.7M training meshes (Figure 10). This multi-source setup covers a diverse range of hand shapes, poses, and object interactions.

The loss function combines full-mesh supervision with an auxiliary loss on the 16 markers retained by the Transformer:

$$\mathcal{L} = \frac{1}{V} \sum_{v=1}^V \|\mathbf{v}_v^{\text{pred}} - \mathbf{v}_v^{\text{gt}}\|_2^2 + \lambda_{\text{markers}} \frac{\sum_{n=1}^{16} \alpha_n \|\mathbf{m}_n^{\text{pred}} - \mathbf{m}_n^{\text{aug}}\|_2^2}{\sum_{n=1}^{16} \alpha_n} \quad (3)$$

where $\mathbf{v}^{\text{pred}}, \mathbf{v}^{\text{gt}}$ are the predicted and ground-truth MANO mesh vertices ($V = 778$), and $\mathbf{m}^{\text{pred}}, \mathbf{m}^{\text{aug}}$ are the retained predicted and augmented markers. We set $\lambda_{\text{markers}} = 0.5$. The fingertip weights α_n are 2.0 and all other retained-marker weights are 1.0.

Optimization. We train for 50 epochs with Adam at a learning rate of 5×10^{-6} , batch size 320, bf16 mixed precision, and gradient clipping at 1.0 on five GPUs. Validation is performed every epoch, and the checkpoint with the lowest validation loss is retained. The run is initialized from the compatible warm-start checkpoint specified in the released configuration.

Data augmentation. To improve robustness to motion-capture variation, we apply the active perturbations specified in `m2m_config.yaml` to every training batch; validation inputs are left unperturbed apart from deterministic root centering.

- **Structural:** (1) Bone-length perturbation—each bone segment is selected independently with probability 0.3 and scaled by up to $\pm 5\%$ of its length, simulating soft-tissue artifacts. (2) Random global scaling—after root-centering, all coordinates are multiplied by a factor in $[0.75, 1.25]$, varying the apparent hand scale.
- **Identity:** (3) Marker swap—spatially proximate markers (within 15 mm) are randomly exchanged with probability 0.3 (capped at three swaps per frame). (4) Neighborhood blending—each marker is blended with a random convex combination of its neighboring mesh vertices, retaining a self-weight of 0.5.
- **Noise:** (5) Gaussian noise—i.i.d. $\mathcal{N}(0, 1 \text{ mm}^2)$ noise is added, with the noise suppressed for a marker with probability 0.1. (6) Marker drift—three markers receive a systematic spatial offset of up to 5 mm in a random direction.

Marker dropout and marker-spike augmentation are disabled in the reported configuration.

To assess the fidelity of markers2mano reconstruction, we report the reconstruction error of the original markers and virtual vertices of the reconstructed meshes. *markers2mano* shows 3.2 mm of per-vertex error on the public dataset, and 4.3 mm of per-marker error on the *EgoEMG*. We also manually reproject meshes onto egocentric RGB frames sampled randomly to validate the label quality, as shown in Figure 11.

Wrist Articulation Angle Computation

Wrist articulation angles are computed directly from motion-capture markers, independent of the MANO reconstruction pipeline. Markers attached to each EMG armband (4 markers on the right wrist, 5 on the left) define a forearm coordinate frame, from which we compute two wrist degrees of freedom: flexion/extension θ_{fe} and radial/ulnar deviation θ_{ru} .

Three non-collinear armband markers define a forearm frame with normalized basis vectors $\hat{\mathbf{F}} = \text{normalize}(\mathbf{F})$ (forearm direction) and $\hat{\mathbf{L}} = \text{normalize}(\mathbf{L})$ (leftward across dorsal surface), with normal $\hat{\mathbf{N}} = \text{normalize}(\hat{\mathbf{F}} \times \hat{\mathbf{L}})$. Let $\hat{\mathbf{H}}$ be the unit vector from wrist to middle finger MCP, and $\hat{\mathbf{H}}_{\text{proj}} = \hat{\mathbf{H}} - (\hat{\mathbf{H}} \cdot \hat{\mathbf{N}})\hat{\mathbf{N}}$ its projection onto the $(\hat{\mathbf{F}}, \hat{\mathbf{L}})$ plane. The wrist angles are:

$$\theta_{\text{fe}} = \arcsin(\hat{\mathbf{H}} \cdot \hat{\mathbf{N}}), \quad \theta_{\text{ru}} = \text{atan2}(\hat{\mathbf{H}}_{\text{proj}} \cdot \hat{\mathbf{L}}, \hat{\mathbf{H}}_{\text{proj}} \cdot \hat{\mathbf{F}}),$$

with extension and radial deviation defined as positive. For the left hand, $\mathbf{N} = \mathbf{L} \times \mathbf{F}$ to maintain consistent sign conventions.

Joint Angle Conversion from MANO to UmeTrack / EMG2Pose Format

In addition to MANO pose and shape parameters (θ, β) , *EgoEMG* provides per-frame joint angles in the 20-DoF scalar format used by EMG2Pose and UmeTrack (Salter et al. 2024; Han et al. 2022), enabling direct comparison with prior EMG-to-pose benchmarks that use scalar joint angle representations. The conversion is formulated as landmark-level inverse kinematics.

Conversion pipeline. Given MANO pose and shape parameters for a frame, we first run MANO forward kinematics to obtain 21 joint positions. We align these into the UmeTrack coordinate frame via a fixed transformation (scale, global orientation, and translation) precomputed by jointly optimizing alignment between the two models’ rest-pose meshes. We then optimize 20 scalar joint angles $\mathbf{a} \in \mathbb{R}^{20}$ such that the UmeTrack forward kinematics produces landmark positions matching the aligned MANO targets:

$$\mathcal{L}(\mathbf{a}) = \frac{1}{20} \sum_{i=1}^{20} \|\mathbf{p}_i^{\text{umetrack}}(\mathbf{a}) - \mathbf{p}_i^{\text{mano}}\|_2^2, \quad (4)$$

where $\mathbf{p}_i^{\text{mano}}$ are the aligned MANO joint positions (after applying the fixed scale, rotation, and translation) and $\mathbf{p}_i^{\text{umetrack}}(\mathbf{a})$ are the corresponding UmeTrack landmarks computed via the UmeTrack forward kinematics (scalar joint angles $\rightarrow$ axis-angle rotations via Rodrigues $\rightarrow$ skinning). The 21 MANO joints are mapped to 20 UmeTrack landmarks via a predefined correspondence; the MANO wrist joint is excluded because it coincides with the UmeTrack coordinate origin and carries no pose information.

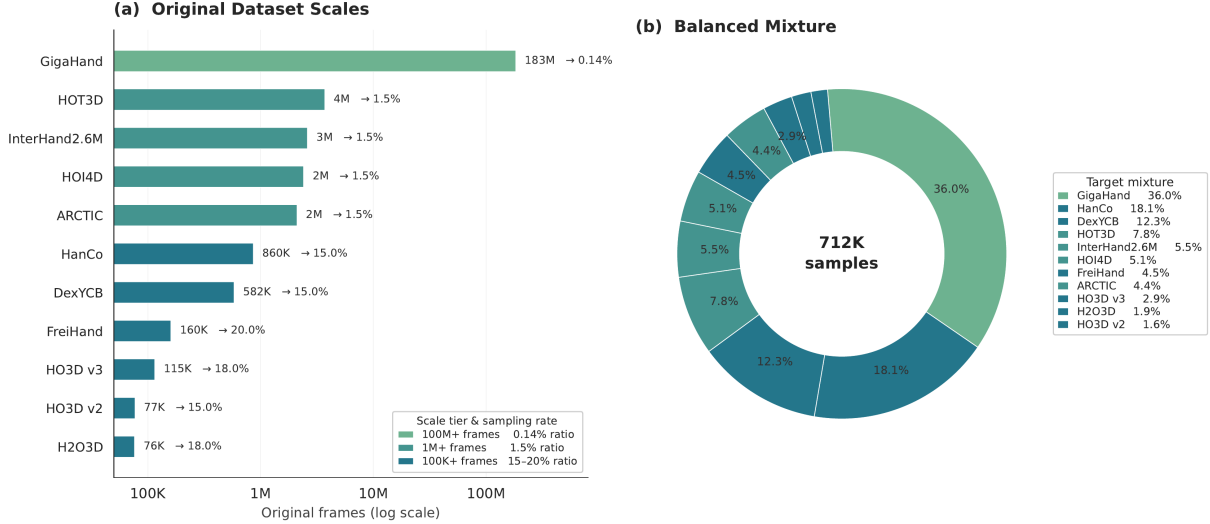


Figure 10: Composition of the *markers2mano* training data. Left: original frame counts of 11 public datasets (log scale), spanning 76K to 183M frames. Each dataset is assigned a sampling ratio by scale tier—0.14% (GigaHand, 100M+), 1.5% (1M+), or 15–20% (100K+)—shown as annotations. Right: target distribution of the training mixture after applying the tiered sampling ratios, yielding approximately 0.7M training meshes.

Optimization. We minimize $\mathcal{L}(\mathbf{a})$ with L-BFGS (strong Wolfe line search, learning rate 0.1, 100 outer steps with up to 50 inner iterations each). Joint range constraints from the UmeTrack hand model are enforced via sigmoid reparameterization:

$$\mathbf{a} = \mathbf{a}_{\min} + (\mathbf{a}_{\max} - \mathbf{a}_{\min}) \odot \sigma(\mathbf{z}),$$

where $\mathbf{z}$ is the unconstrained optimizable variable and $\odot$ denotes element-wise multiplication. The optimization runs per-episode on GPU with batch size 256; episodes exceeding GPU memory are processed in smaller chunks.

Output. The resulting 22-DoF prediction target combines 20 finger joint angles (indices 0–19) and 2 wrist articulation angles (indices 20–21). The mean per-landmark fitting error of the optimized UmeTrack joints relative to the MANO targets is 2.8 mm, confirming that the conversion introduces negligible additional error. Table 6 lists the complete semantic ordering per dimension. The conversion script is included in the dataset release.

Table 6: Complete 22-DoF joint angle target semantics. Each dimension is reported in degrees. FE = flexion/extension, AA = abduction/adduction. Indices 0–3 cover the thumb, 4–7 the index finger, 8–11 the middle finger, 12–15 the ring finger, and 16–19 the pinky.

Index	Finger	Joint	Motion
0	Thumb	CMC	FE
1	Thumb	CMC	AA
2	Thumb	MCP	FE
3	Thumb	IP	FE
4	Index	MCP	AA
5	Index	MCP	FE
6	Index	PIP	FE
7	Index	DIP	FE
8	Middle	MCP	AA
9	Middle	MCP	FE
10	Middle	PIP	FE
11	Middle	DIP	FE
12	Ring	MCP	AA
13	Ring	MCP	FE
14	Ring	PIP	FE
15	Ring	DIP	FE
16	Pinky	MCP	AA
17	Pinky	MCP	FE
18	Pinky	PIP	FE
19	Pinky	DIP	FE
20	–	Wrist	Flexion/Extension
21	–	Wrist	Radial/Ulnar deviation

We show reconstruction samples in Figure 11.

Training and implementation details

EMGFormer Training Details

Optimization. For the reported *EgoEMG* results, all EMGFormer variants use target-wrist 8-channel EMG windows of 12,000 samples (6 s) and are trained for 150 epochs. We use AdamW (Loshchilov and Hutter 2019) ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2}), an initial learning rate of 5×10^{-4} with cosine annealing to 5×10^{-6} , gradient clipping at 1.0, and bf16 mixed precision. The objective is

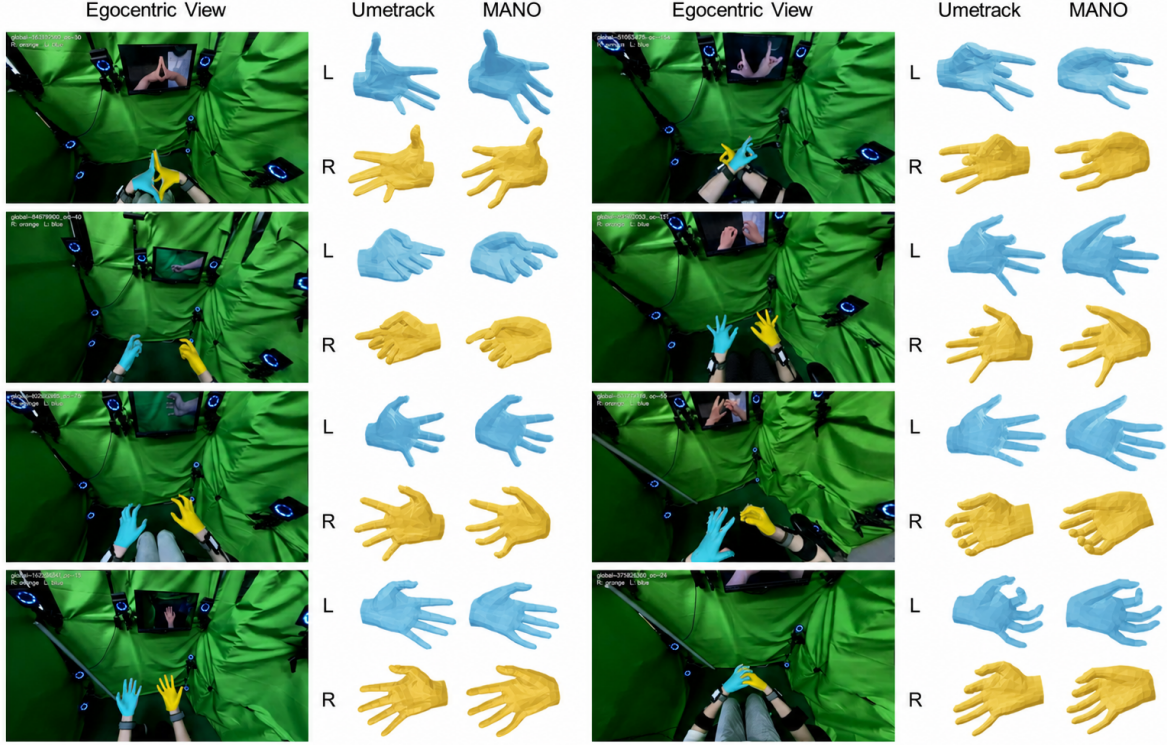


Figure 11: Mesh reprojection and visualization of UmeTrack and MANO mesh reconstruction on *EgoEMG*. The reconstructed MANO meshes (from *markers2mano*) are overlaid on egocentric RGB frames, demonstrating accurate hand pose reconstruction and realistic mesh-to-image alignment.

Table 7: Architecture and training configuration per EMGFormer variant. “Params” includes the TDS encoder, Transformer decoder, and regression head.

Variant	L	d	Heads	d_{FFN}	Params
EMGFormer-S	3	256	4	512	3.5M
EMGFormer-M	6	256	8	1024	6.6M
EMGFormer-L	8	384	12	1536	16.3M

L1 joint-angle loss with a fingertip-distance term of weight 0.01. Per-dataset normalization statistics are computed from the training data and then fixed for validation and testing. Training uses four to six NVIDIA RTX 4090 GPUs depending on model size.

Data augmentation. We apply the same on-the-fly augmentation family to all three sizes: random gain, magnitude warping, baseline drift, power-line noise, time/frequency masking, additive Gaussian noise, channel masking, mixup, and temporal window jitter. Exact probabilities and ranges are included in the released experiment configuration.

Architecture and per-size configuration. Table 7 summarizes the architecture and training configuration per EMGFormer variant.

TDS featurizer. The TDS-style temporal convolutional frontend (Hannun et al. 2019) maps a filtered 2 kHz EMG window ($T = 12,000$ samples, 8 target-wrist channels) to a 256-dimensional feature sequence. The architecture consists of two initial strided Conv1d blocks followed by two TDS stages. Each TDS stage begins with a strided Conv1d for temporal subsampling, followed by a TDSCovEncoder composed of alternating depth-separable temporal convolutions (reshaping the channel axis into a 2D feature map) and channel-wise fully-connected blocks. Squeeze-and-excitation (Hu, Shen, and Sun 2018) is applied globally after each TDS stage. The layer-wise configuration is:

1. Conv1d: $8 \rightarrow 256$, kernel 11, stride 5
2. Conv1d: $256 \rightarrow 256$, kernel 5, stride 2
3. TDS Stage 1: in-conv kernel 9, stride 5; 1 block of TDSCov (kernel width 5, channels 8, feature width 32)
4. TDS Stage 2: in-conv kernel 3, stride 1; 1 block of TDSCov (kernel width 3, channels 8, feature width 32)

The final TDS output is a $(256, 230)$ feature map, corresponding to an effective frame rate of approximately 38 Hz. The subsequent non-causal Transformer contextualizes this sequence over the full input window.

Vision and Fusion Training Details

Vision baseline. We provide six fully fine-tuned generic vision backbones: ResNet- $\{18, 50, 152\}$ and ViT- $\{\text{Small, Base, Large}\}$. ResNet backbones are initialized from ImageNet-

Table 8: Vision backbone configuration. All models take 256×256 center-frame crops as input and are evaluated on the same 22-DoF target. Generic ResNet/ViT models share the same MLP prediction head; WiLoR uses its native hand-pose backbone with a 22-DoF regression head.

Backbone	Params	d_{backbone}	LR	Batch size	Epochs
ResNet-18	11.5M	512	10^{-3}	900	150
ResNet-50	23.5M	2048	10^{-3}	400	150
ResNet-152	58.2M	2048	10^{-3}	180	150
ViT-S/14	21.9M	384	10^{-4}	360	150
ViT-B/14	86.2M	768	10^{-4}	160	150
ViT-L/14	303.8M	1024	10^{-4}	50	150
WiLoR	631.6M	1280	10^{-5}	18	30

pretrained weights, while ViT backbones are initialized from DINOv2-pretrained weights. Each generic backbone extracts a feature vector from the window center frame crop (256×256), which is fed through a prediction head to regress 22 joint angles. The prediction head is a 2-layer MLP: $\text{Linear}(d_{\text{backbone}} \rightarrow 512) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.1) \rightarrow \text{Linear}(512 \rightarrow 22)$. All generic vision models are trained with the same L1 (MAE) loss plus fingertip-distance loss (weight 0.01) as the EMG baseline. We additionally fine-tune WiLoR (Potamias et al. 2024) as a hand-specialized baseline using its pretrained MANO-based checkpoint, with the same center-frame supervision target and output dimensionality. Generic backbones use AdamW (Loshchilov and Hutter 2019) with cosine annealing and bf16 mixed precision on 6 NVIDIA RTX 4090 GPUs; WiLoR uses AdamW with learning rate 10^{-5} and cosine annealing over 30 epochs. Table 8 lists the per-backbone configuration.

Fusion baseline. The fusion model uses the residual architecture described in the main paper. The vision pathway (backbone, projection layer, and vision head) is initialized from the corresponding vision-only checkpoint, and the EMG pathway (EMGFormer-S, comprising a TDS featurizer and a Transformer decoder) is initialized from an EMG-only checkpoint. The fusion projection and residual regression head are randomly initialized, with the last layer set to zero so that the initial prediction matches the vision-only baseline ($\Delta y_{\text{emg}} \approx 0$). The model predicts only the center frame of the window and is trained with AdamW (Loshchilov and Hutter 2019), cosine learning-rate annealing, the same L1 plus fingertip-distance objective, and bf16 mixed precision. We adjust batch size to backbone memory requirements. Exact per-run learning rates, cosine endpoints, epoch budgets, batch sizes, sampling strides, dataset-repeat factors, augmentation settings, and visual-update strategies are provided in the accompanying Hydra configurations.

Extended experimental results

Frozen vs. Fine-Tuned Vision in Fusion

Table 9 reports the complete 2×7 comparison underlying the model selection in Table 4. The paired variants use the same residual-fusion architecture, pretrained single-modality

initializers, and center-frame supervision, while the visual predictor is either frozen or jointly fine-tuned. Backbone-specific optimization and sampling settings are recorded in the accompanying experiment configurations. Selection is performed independently for each backbone using validation MAE.

Per-Hand Breakdown of EMG Baseline Results

Table 10 reports per-hand MAE for representative methods on *EgoEMG*, complementing the aggregate results in Table 2.

Left and right hand MAE agree within 2.0° across all models and splits, confirming that the shared-weight bilateral design and left-to-right mirroring strategy produce symmetric performance. The per-hand consistency supports the validity of processing both hands through a single model.

Per-Joint Error Analysis

Table 11 presents the per-finger and per-joint-group mean absolute error (MAE) for EMGFormer variants on *EgoEMG*. Across all three model sizes, distal joints and wrist deviation yield the lowest errors (best MAE 10.0° for both), whereas the pinky and mid-phalanx groups are the most challenging (best MAE 16.7° and 20.7° , respectively). This pattern is consistent with hand kinematics and sensing difficulty. Distal joints tend to have a more constrained range of motion, making their angles easier to regularize from EMG signals. In contrast, mid-phalanx joints capture a larger portion of finger flexion and therefore exhibit greater motion variability. The higher error on the pinky is also expected: pinky motion is often weaker, more coupled with the ring finger, and less independently represented in wrist-level EMG, making it harder to distinguish from neighboring-finger activation patterns.

For comparison, Table 12 details the corresponding per-joint MAE on the EMG2Pose dataset. Because EMG2Pose lacks wrist articulation labels, our evaluation here is restricted to finger joints.

A cross-dataset comparison shows that mid-phalanx and pinky joints are consistently among the most challenging groups across both benchmarks. However, the easiest groups differ between datasets: proximal joints achieve the lowest MAE on EMG2Pose, whereas distal joints and wrist deviation are easiest on *EgoEMG*. The overall error magnitudes are also higher on *EgoEMG*—ranging from 10.0° to 21.6° , compared with 8.5° to 14.2° on EMG2Pose—which likely reflects differences in participant count, gesture diversity, sensor setup, and the inclusion of wrist articulation in *EgoEMG*’s 22-DoF prediction target.

Per-Gesture Modality Complementarity Analysis

To characterize where EMG provides complementary information beyond vision, we evaluate all three modalities—vision-only, EMG-only, and Fusion-RN18+S—on a per-gesture basis across the full test set (all splits, both hands). Following the Gesture Vocabulary section, we group the 60 gesture classes into three semantic families: single-hand (30 gestures), symmetric bimanual (18), and asymmetric bimanual (12). One bimanual gesture (raw, unconstrained) is excluded from this

Table 9: Complete frozen-versus-fine-tuned vision ablation for EMG+vision fusion. Gesture/User/Both report per-subject mean $\pm$ standard deviation in degrees; Avg is the mean over all splits (lower is better). Shading marks the variant selected by validation MAE.

Backbone	Update	Gesture	User	Both	Avg
ResNet-18	Frozen	5.04 $\pm$ 1.25	6.74 $\pm$ 0.33	6.64 $\pm$ 0.79	5.83
	Fine-tuned	4.60 $\pm$ 1.30	6.40 $\pm$ 1.40	6.40 $\pm$ 0.80	5.40
ResNet-50	Frozen	4.39 $\pm$ 1.18	6.19 $\pm$ 0.23	6.05 $\pm$ 0.60	5.19
	Fine-tuned	4.47 $\pm$ 1.23	6.20 $\pm$ 0.25	6.10 $\pm$ 0.59	5.24
ResNet-152	Frozen	4.25 $\pm$ 1.17	6.04 $\pm$ 0.27	6.03 $\pm$ 0.59	5.06
	Fine-tuned	4.34 $\pm$ 1.18	6.11 $\pm$ 0.32	6.05 $\pm$ 0.64	5.13
ViT-S/14	Frozen	5.01 $\pm$ 1.28	7.33 $\pm$ 0.27	7.13 $\pm$ 0.32	6.02
	Fine-tuned	4.50 $\pm$ 1.30	6.80 $\pm$ 1.20	6.80 $\pm$ 0.60	5.54
ViT-B/14	Frozen	4.80 $\pm$ 1.26	7.03 $\pm$ 0.33	6.84 $\pm$ 0.41	5.75
	Fine-tuned	4.79 $\pm$ 1.27	7.03 $\pm$ 0.32	6.87 $\pm$ 0.42	5.76
ViT-L/14	Frozen	4.34 $\pm$ 1.26	6.69 $\pm$ 0.36	6.60 $\pm$ 0.56	5.36
	Fine-tuned	4.32 $\pm$ 1.27	6.69 $\pm$ 0.35	6.63 $\pm$ 0.57	5.37
WiLoR	Frozen	3.96 $\pm$ 1.21	5.52 $\pm$ 0.42	5.54 $\pm$ 0.62	4.68
	Fine-tuned	3.95 $\pm$ 1.21	5.54 $\pm$ 0.39	5.54 $\pm$ 0.59	4.69

Table 10: Per-hand EMG-to-pose MAE on *EgoEMG*. MAE is reported in degrees; lower is better. Left and right hands are processed independently with shared model weights, and left-hand EMG is mirrored to the right-hand convention. Rows are grouped by method family.

Method	Hand	Gesture	User	Both
<i>EMGFormer variants</i>				
EMGFormer-S	Left	11.8	15.8	16.7
	Right	12.6	16.5	16.7
EMGFormer-M	Left	11.3	15.8	16.6
	Right	12.0	16.4	16.9
EMGFormer-L	Left	11.3	15.7	16.5
	Right	12.4	16.6	16.8
<i>Traditional architectures</i>				
vemg2pose	Left	14.7	15.7	17.2
	Right	15.2	17.0	17.5
NeuroPose	Left	15.9	15.4	15.7
	Right	15.7	16.0	16.9

analysis due to its unbounded pose distribution. Table 13 reports sample-weighted per-family averages across all three modalities.

Three patterns emerge. First, the fusion gain is broadly distributed: fusion improves over vision-only on 54 of 59 analyzed gesture classes, suggesting that the aggregate improvement is not confined to only a few gesture classes. Second, the largest family-level gain occurs on single-hand gestures ($\Delta = +0.50^\circ$), followed by asymmetric bimanual ($\Delta = +0.41^\circ$). This suggests that EMG is most useful when

Table 11: Per-joint MAE ($^\circ$) on *EgoEMG*, left and right hands pooled within each split, then per-sample weighted average across the Gesture, User, and Both splits. Lower is better. Bold indicates the best mean per row.

Joint group	EMGFormer-S	EMGFormer-M	EMGFormer-L
Distal	10.2	10.1	10.0
Wrist deviation	10.8	10.1	10.0
Thumb	12.8	12.3	12.3
Index	13.4	13.3	13.3
Proximal	13.7	13.3	13.4
Ring	15.2	14.7	14.6
Middle finger	15.2	14.8	14.8
Wrist flexion	15.7	14.9	14.6
Pinky	17.3	16.7	17.0
Mid-phalanx	21.6	20.7	20.7

the prediction depends on subtle within-hand articulation or asymmetric cross-hand coordination that is harder to infer from a single RGB crop. Symmetric bimanual gestures show the smallest gain ($\Delta = +0.38^\circ$), as two-hand interactions of this type are often easier to discriminate visually. Third, even for the few gestures where fusion slightly underperforms vision-only (Rock, -0.22° ; Typing, -0.21° ; Clap, -0.12° ; FingerPullLeft, -0.04° ; CrossHand, -0.01°), the degradation is negligible and involves large-scale hand silhouettes whose visual features are already highly discriminative. Table 14 provides the complete per-gesture breakdown for all 59 analyzed gesture classes, grouped by semantic family and sorted by descending fusion gain within each family.

Table 12: Per-joint MAE ($^{\circ}$) on the EMG2Pose dataset, per-sample weighted average across all three test splits (stage, user, and user+stage). Because EMG2Pose lacks wrist labels, only finger joints are reported. Bold indicates the best mean per row.

Joint group	EMGFormer-S	EMGFormer-M	EMGFormer-L
Proximal	9.3	8.9	8.5
Index	10.8	10.3	9.8
Thumb	11.1	10.6	9.9
Middle finger	11.1	10.5	10.0
Ring	11.1	10.6	10.1
Distal	13.3	12.8	12.2
Pinky	13.5	12.9	12.3
Mid-phalanx	14.2	13.4	12.6

Table 13: Per-gesture-family modality comparison on *EgoEMG* (Fusion-RN18+S). Values are sample-weighted averages across all test splits; MAE is in degrees (lower is better). $\Delta = \text{MAE}_{\text{vision}} - \text{MAE}_{\text{fusion}}$; positive Δ indicates fusion improvement. Gesture families follow the Gesture Vocabulary section.

Gesture family	#G	#Samp.	Vis.	EMG	Fusion	Δ
Single-hand	30	2,406	6.0	15.0	5.5	+0.50
Symmetric bimanual	17	1,288	5.9	14.0	5.5	+0.38
Asymmetric bimanual	12	584	6.8	14.6	6.4	+0.41
Overall	59	4,278	6.1	14.6	5.6	+0.46

Fusion Benefit vs. Hand Self-Occlusion

To understand when EMG–vision fusion provides the greatest benefit, we stratify test-set predictions by the degree of hand self-occlusion.

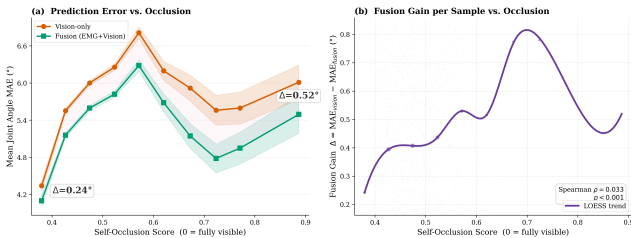


Figure 12: Fusion benefit versus hand self-occlusion on *EgoEMG*. (a) Vision-only and EMG+vision MAE across 0.05-wide occlusion bins. (b) Per-bin fusion gain ($\Delta = \text{MAE}_{\text{vision}} - \text{MAE}_{\text{fusion}}$) with a LOESS trend. Error bands denote ± 1 SEM.

Self-occlusion score. We quantify per-frame hand self-occlusion via z-buffer rasterization of the ground-truth MANO mesh in camera space. Concretely, for each test frame:

1. The MANO mesh vertices are transformed from world coordinates to camera coordinates using the calibrated extrinsic matrix ($\mathbf{v}_{\text{cam}} = \mathbf{R}_{C \leftarrow W} \mathbf{v}_W + \mathbf{t}_{C \leftarrow W}$).
2. All mesh triangles are rasterized into a depth buffer (z-buffer) at the native video resolution via pinhole projection

with the calibrated intrinsic matrix $\mathbf{K}$. For each pixel covered by a triangle, the interpolated depth is stored if it is smaller than the current buffer value.

3. A vertex is deemed *visible* if, within a 5×5 pixel neighbourhood of its projected location, any z-buffer entry agrees with the vertex depth to within $\epsilon = 5$ mm.
4. Each triangle’s 3-D surface area (half the cross-product norm of two edge vectors) is distributed equally to its three vertices, yielding a per-vertex area weight w_i .
5. The **self-occlusion score** is defined as

$$s_{\text{occ}} = 1 - \frac{\sum_{i \in \mathcal{V}_{\text{vis}}} w_i}{\sum_{i=1}^V w_i}, \quad (5)$$

where $\mathcal{V}_{\text{vis}}$ is the set of visible vertices. A score of 0 means the hand is fully visible; a score of 1 means it is fully occluded from the camera viewpoint.

Results. Figure 12(a) shows that both vision-only and fusion MAE vary with occlusion, while the gap between them—the fusion gain—is consistently positive across the full range, from $\Delta = 0.24^{\circ}$ at the lowest occlusion bin to $\Delta = 0.52^{\circ}$ at the highest. The largest fusion benefit occurs at moderate-to-high occlusion ($\Delta \approx 0.8^{\circ}$ near occlusion ≈ 0.72), where visual evidence is substantially degraded. Panel (b) visualizes the same trend at the per-bin level with a LOESS smooth. The frame-level association between occlusion and fusion gain is positive but weak in magnitude (Spearman $\rho = 0.033$, $p < 0.001$; 16,650 valid predictions), so we interpret the bin-wise pattern as suggestive rather than as evidence of a strong monotonic relationship.

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Table 14: Full per-gesture modality breakdown on *EgoEMG* using Fusion-RN18+S. MAE is in degrees. $\Delta = \text{MAE}_{\text{vis}} - \text{MAE}_{\text{fus}}$. Gestures are grouped by family and sorted by descending Δ within each family. Gesture descriptions are provided in the Gesture Vocabulary section.

ID	Name	N	Vis.	EMG	Fus.	Δ
<i>Single-hand gestures</i>						
6	ASL7	36	7.2	16.1	6.2	+0.96
8	ASL9	38	5.7	17.5	5.0	+0.71
12	ILY	40	7.3	19.2	6.6	+0.70
13	IndexBow	34	6.3	20.7	5.6	+0.66
20	PinchMiddle	324	5.3	12.2	4.7	+0.64
26	Thumb	40	7.9	24.8	7.3	+0.62
7	ASL8	42	7.0	14.8	6.4	+0.61
28	nocontact_free	292	6.8	14.2	6.2	+0.61
18	Nine	320	7.0	18.1	6.4	+0.59
17	MiddleBow	92	4.1	11.1	3.5	+0.57
16	JoystickSlide	38	5.6	19.5	5.0	+0.56
23	RingAndThumb	36	7.2	16.5	6.7	+0.55
9	Claw3	38	7.7	17.3	7.2	+0.54
27	nocontact_disperse_palm	92	4.9	12.2	4.5	+0.46
15	JoystickCircle	308	4.5	12.2	4.0	+0.45
2	ASL3	30	7.3	19.6	6.8	+0.44
11	FreeAction	38	6.5	15.8	6.1	+0.41
5	ASL6	40	6.9	18.1	6.5	+0.39
14	IndexMiddleClaw	92	5.0	12.9	4.6	+0.39
1	ASL2	34	6.9	17.7	6.5	+0.38
19	PalmYaw	82	3.2	9.8	2.8	+0.36
4	ASL5	28	4.5	14.6	4.2	+0.35
24	RingBow	38	7.3	16.9	7.0	+0.34
0	ASL1	44	7.9	17.6	7.7	+0.26
10	Claw5	36	6.0	15.6	5.7	+0.25
22	Rest	30	7.1	17.4	6.9	+0.20
21	PinkyBow	40	7.5	16.6	7.4	+0.18
3	ASL4	32	6.2	18.0	6.1	+0.14
29	nocontact_grab	36	7.4	18.6	7.4	+0.06
25	Rock	36	9.5	17.5	9.7	-0.22
<i>Symmetric bimanual gestures</i>						
38	HandClasp	30	8.7	16.4	8.0	+0.75
52	MiddleOppo	226	5.3	12.3	4.8	+0.59
39	HandRub	32	5.9	16.0	5.3	+0.55
36	FistBump	256	9.2	23.7	8.7	+0.51
44	Prayer	78	3.9	9.5	3.4	+0.46
42	PalmStack	262	4.6	10.2	4.2	+0.41
47	SymSwing	28	6.6	15.7	6.2	+0.36
40	IndexTapping	32	5.3	13.0	5.0	+0.30
37	Gaming	26	7.8	15.9	7.5	+0.24
46	SymOpen	30	5.0	12.6	4.8	+0.22
51	Kiss	78	4.0	10.4	3.7	+0.21
48	ThumbWrestle	34	5.5	14.6	5.3	+0.18
35	FingerTipTouch	76	3.6	7.5	3.4	+0.16
32	CrossStretch	26	5.3	11.9	5.3	+0.01
31	CrossHand	26	5.6	12.8	5.7	-0.01
30	Clap	14	6.7	14.3	6.8	-0.12
49	Typing	34	5.3	11.9	5.5	-0.21
<i>Asymmetric bimanual gestures</i>						
34	FingerPullRight	28	7.1	13.6	6.1	+0.96
58	PinchWring	32	7.5	17.8	6.8	+0.71
55	PairClaw	36	8.1	17.4	7.4	+0.66
43	PinkyHook	32	7.0	16.0	6.5	+0.51
53	Beijing	30	10.1	19.2	9.6	+0.44
41	PalmRoll	54	4.4	12.1	4.0	+0.36
59	ThumbOppo	30	7.1	17.5	6.7	+0.34
45	Squeeze	230	6.2	12.4	5.8	+0.32
54	Checky	22	9.6	21.1	9.4	+0.26
56	PairOK	32	7.6	15.6	7.5	+0.07
57	Picture	32	7.1	16.7	7.1	+0.03
33	FingerPullLeft	26	6.0	13.6	6.0	-0.04

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